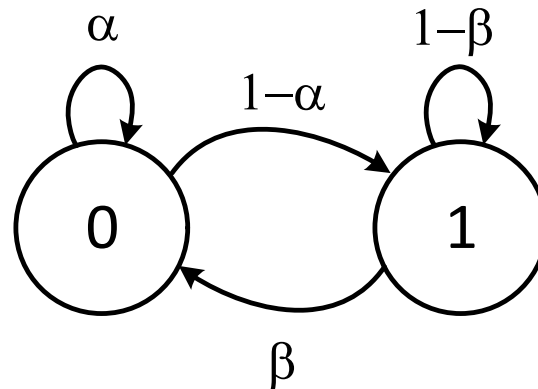
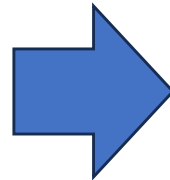


Mathematics of DTMCs - summary

Example: 2-DTMC



$$P = \begin{bmatrix} 0.1 & 0.9 \\ 0.6 & 0.4 \end{bmatrix}$$



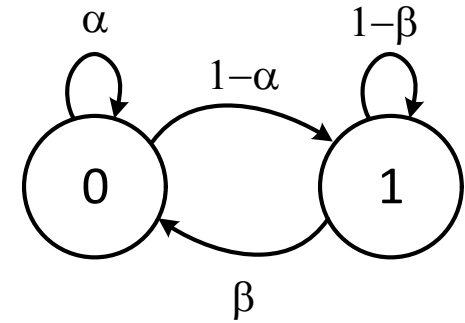
$$P^{(n)} = P^n$$

$$P^{(3)} = P^3 = \begin{bmatrix} 0.325 & 0.675 \\ 0.45 & 0.55 \end{bmatrix}$$

(one-step)
transition probability matrix

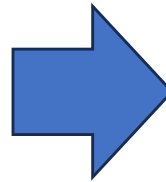
n -step
transition probabilities matrix

Mathematics of DTMCs - summary



$$\mathbf{P}^{(n)} = \mathbf{P}^n$$

$$\mathbf{P}^{(3)} = \mathbf{P}^3 = \begin{bmatrix} 0.325 & 0.675 \\ 0.45 & 0.55 \end{bmatrix}$$

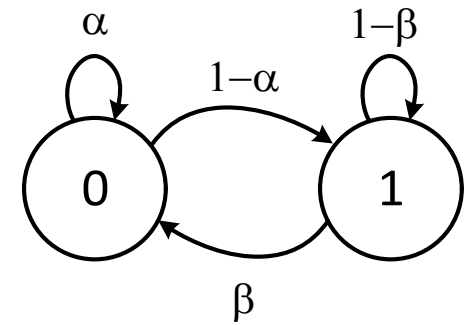


$$P(X_n = j) = \alpha_j^n = \sum_{i=0}^{N-1} P_{ij}^n \alpha_i$$

n -step
transition probabilities matrix

transient state probabilities

Mathematics of DTMCs - summary



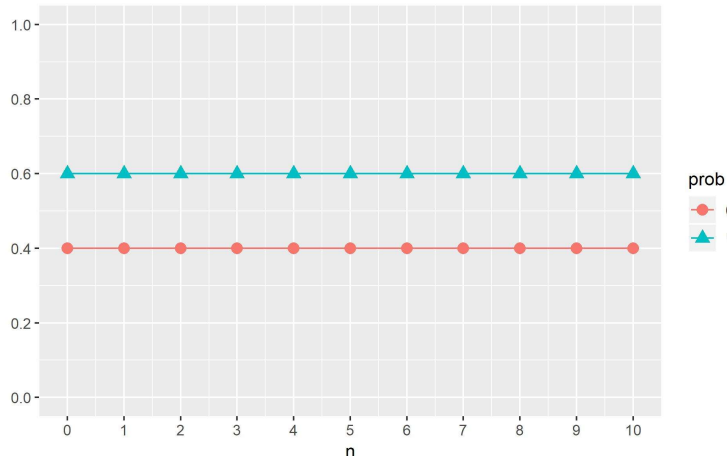
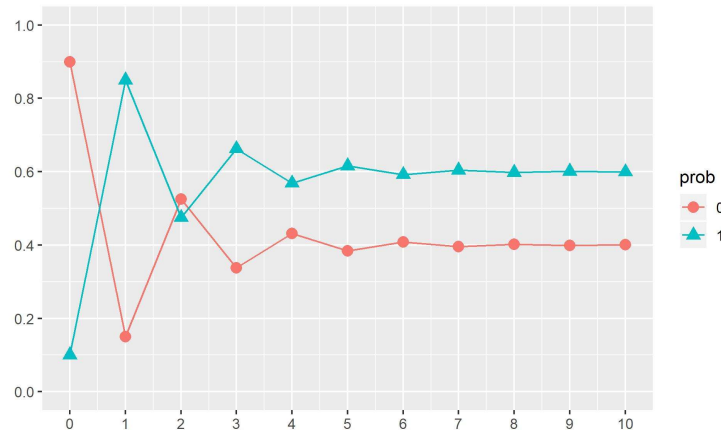
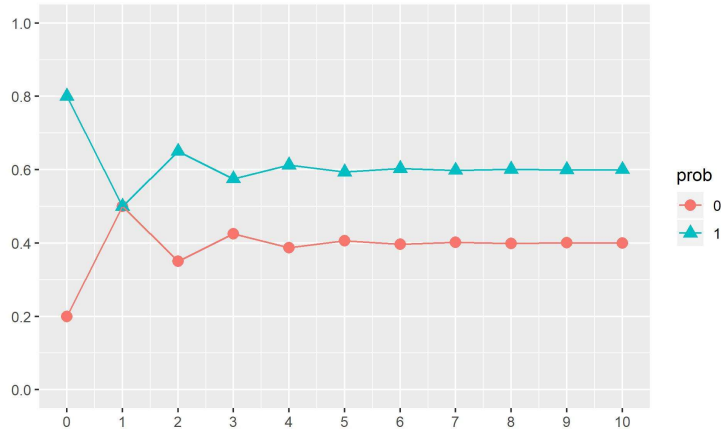
$$P(X_n = j) = \alpha_j^n = \sum_{i=0}^{N-1} P_{ij}^n \alpha_i$$

transient state probabilities

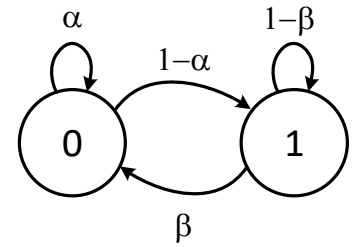


$$\pi_j^L = \lim_{n \rightarrow \infty} \alpha_j^n = \lim_{n \rightarrow \infty} P_{ij}^n$$

Limiting state probabilities
independent of initial conditions



Mathematics of DTMCs - summary



$$P = \begin{bmatrix} 0.1 & 0.9 \\ 0.6 & 0.4 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.55 & 0.45 \\ 0.3 & 0.7 \end{bmatrix}$$

$$P^5 = \begin{bmatrix} 0.3831 & 0.6188 \\ 0.4125 & 0.5875 \end{bmatrix}$$

$$P^{13} = \begin{bmatrix} 0.3999 & 0.6001 \\ 0.4000 & 0.6000 \end{bmatrix}$$

matrix multiplication
(multiply until column values get
approximately equal)

$$\sum_{i=0}^{N-1} \pi_i^S P_{ij} = \pi_j^S, \quad j = 0, 1, \dots, N-1$$

$$\sum_{i=0}^{N-1} \pi_i^S = 1$$

stationary state probabilities
(equal to the limiting state probabilities, under certain conditions)

$$\pi_j^S \sum_i P_{ji} = \sum_i \pi_i^S P_{ij}, \quad j = 0, 1, \dots, N-1$$

$$\sum_{i=0}^{N-1} \pi_i^S = 1$$

global balance equations
(frequency of transitions out of state = frequency of transition into state)

ways of computing the limiting state probabilities